

TE Curve Verification Pipeline Official Model Verification Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Verdict

TE Curve Verification Pipeline is the official offline verification surface for newly introduced TE models. A model or family is not considered verified only because its training campaign metrics improved; it must also be compared on the direction-aware TE Curve Verification Pipeline curve-reconstruction matrix and reviewed against visual curve evidence.

Current closeout verdict after the wave 2.1 temporal refresh:

- TE Curve Verification Pipeline remains the canonical offline model-verification report.
- The wave 2.1 temporal candidates are verified in the matrix and visual companion reports.
- No temporal candidate is promoted over the current repository-owned tree / hist_gradient_boosting offline baseline.
- The strongest temporal forward candidate is gru_sequence_Fw .
- The strongest temporal backward candidate is lstm_sequence_Bw .
- The strongest temporal global candidate is lstm_sequence_global .
- Future work should treat the temporal models as verified exploratory baselines, not as the accepted offline winner.

Source Package

This official report consolidates these approved `TE Curve Verification Pipeline` artifacts:

- metric matrix:
`doc/reports/analysis/track2/Track 2 Directional Model Comparison.md` ;
- best-model collage report:
`doc/reports/analysis/track2/best_model_collage_report/[2026-05-24]/track2_best_model_collage_report.md` ;
- best-model collage PDF:
`doc/reports/analysis/track2/best_model_collage_report/[2026-05-24]/track2_best_model_collage_report.pdf` ;
- multi-model curve comparison report:
`doc/reports/analysis/track2/multi_model_curve_comparison_report/[2026-05-24]/track2_multi_model_curve_comparison_report.md` ;
- multi-model curve comparison PDF:
`doc/reports/analysis/track2/multi_model_curve_comparison_report/[2026-05-24]/track2_multi_model_curve_comparison_report.pdf` . Machine-readable and visual validation outputs are retained under:
 - `output/validation_checks/track2_reference_comparison/2026-05-24-22-01-57_track2_full_directional_family_matrix_wave2_temporal_refresh/` ;
 - `output/validation_checks/track2_best_model_collage_report/2026-05-24-22-50-30_track2_best_model_collage_report/` ;
 - `output/validation_checks/track2_multi_model_curve_comparison_report/2026-05-24-23-28-22_track2_multi_model_curve_comparison_report/` .

Verification Rule

The repository treats direction as a first-class verification surface:

Surface	Training or Archive Scope	Evaluation Scope
<code>global</code>	forward and backward together	both directions, reported separately
<code>Fw / forward</code>	forward only	forward curves only
<code>Bw / backward</code>	backward only	backward curves only

The rule applies to paper-reference models, `RCIM Model-Bank Reproduction` faithful archives, retuned reference archives, `Wave 1` exported models, `Wave 2.1` temporal registry models, and future `TE Curve Verification Pipeline` campaign candidates.

Pipeline Coverage

Pipeline or Source	TE Curve Verification Pipeline Role	Current Status	Verification Artifact
recovered original RCIM archive	paper-original forward reference	included	directional matrix
retuned RCIM archive	current paper-derived forward and backward baseline	included	directional matrix and visual reports
<code>RCIM Model-Bank Reproduction</code> exact paper-faithful bank	source-faithful reproduction evidence	included	directional matrix and visual reports
<code>Wave 1</code> exported static baselines	repository-owned static candidates	included	directional matrix and visual reports

Pipeline or Source	TE Curve Verification Pipeline Role	Current Status	Verification Artifact
periodic_mlp explicit harmonic campaign	latest family-registry refresh	included in visual refresh	collage and overlay reports
Wave 2.1 temporal entry campaign	temporal convolution, GRU , and LSTM	included	matrix, collage, overlay, PDF
future TE Curve Verification Pipeline campaigns	new verification candidates	append here when approved	matrix, collage, overlay, PDF

Current Numeric Baselines

Best Composite References

Candidate	Source	Direction	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
paper_original_best_Fw	rcim_original	forward	0.002769	0.002951	6.250	13.827
paper_retuned_best_Fw	rcim_retuned	forward	0.001839	0.002041	4.109	9.866
track1_best_Fw	rcim_track1	forward	0.003014	0.003204	6.819	11.638
paper_retuned_best_Bw	rcim_retuned	backward	0.003675	0.004284	7.572	15.645
track1_best_Bw	rcim_track1	backward	0.005027	0.005212	11.860	48.106

Repository-Owned Static Baselines

Direction	Current Strongest Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
forward	tree_Fw	0.003053	0.003395	6.731	11.995
backward	tree_Bw	0.003258	0.003651	7.051	14.116
global, forward side	tree_global	0.002998	0.003364	6.590	11.601
global, backward side	tree_global	0.003290	0.003702	7.118	13.703

Wave 2.1 Temporal Candidates

Direction	Strongest Temporal	Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
forward	gru_sequence_Fw		0.003330	0.003762	7.378	13.029
backward	lstm_sequence_Bw		0.003555	0.003985	7.767	14.507
global, combined	lstm_sequence_global		0.003480	0.003903	7.654	12.430

The temporal family is competitive with the non-tree neural/static baselines, but it does not beat the accepted tree baseline in either direction.

Visual Verification Evidence

Best-Model Collage Evidence

The refreshed best-model collage report now includes:

- forward and backward composite reference collages;
- Wave 1 family-best collages;
- explicit-harmonic periodic_mlp refresh collages;
- Wave 2.1 temporal collages for temporal_convolution , gru_sequence , and lstm_sequence across global , Fw , and Bw . The PDF companion is: doc/reports/analysis/track2/best_model_collage_report/[2026-05-24]/track2_best_model_collage_report.pdf Representative evidence from the refreshed validation output is stored at:
- output/validation_checks/track2_best_model_collage_report/2026-05-24-22-50-30__track2_best_model_collage_report/collages/forward_wave2/gru_sequence_fw.png ;
- output/validation_checks/track2_best_model_collage_report/2026-05-24-22-50-30__track2_best_model_collage_report/collages/backward_wave2/lstm_sequence_bw.png .

Multi-Model Overlay Evidence

The refreshed multi-model curve comparison report now includes dedicated wave 2.1 overlays and combined reference/tree/temporal overlays for both directions. The combined overlays show the temporal models against the current paper-derived references, RCIM Model-Bank Reproduction references, and tree baseline on the same curves.

The PDF companion is doc/reports/analysis/track2/multi_model_curve_comparison_report/[2026-05-24]/track2_multi_model_curve_comparison_report.pdf

Representative evidence from the refreshed validation output is stored at:

- output/validation_checks/track2_multi_model_curve_comparison_report/2026-05-24-23-28-22__track2_multi_model_curve_comparison_report/comparisons/forward_reference_tree_wave2.png ;
- output/validation_checks/track2_multi_model_curve_comparison_report/2026-05-24-23-28-22__track2_multi_model_curve_comparison_report/comparisons/backward_reference_tree_wave2.png .

Campaign Update Ledger

Future TE Curve Verification Pipeline campaigns must append a row here after the campaign result report, matrix refresh, visual report refresh, and official PDF validation are complete.

Date	Campaign or Update	Candidate Scope	Matrix Status	Visual Status	Decision
2026-05-21	periodic_mlp explicit harmonic registry refresh	global , Fw , Bw visual verification	source matrix unchanged; visual registry refresh included	collage and overlay PDFs refreshed	included as visual evidence, not promoted over tree
2026-05-24	Wave 2.1 temporal-model entry campaign	global , Fw , Bw for TCN, GRU , and LSTM	matrix refreshed from 75 to 84 candidates	collage and overlay PDFs refreshed	verified, but not promoted over tree

Maintenance Contract

For every future `TE Curve Verification Pipeline` model-verification update:

1. refresh `Track 2 Directional Model Comparison.md` when the candidate set or numeric matrix changes;
2. regenerate the best-model collage report when a new family or promoted candidate needs local curve inspection;
3. regenerate the multi-model curve comparison report when relative visual screening changes;
4. append the campaign or update to this report ledger;
5. export and validate this official report PDF;
6. update `doc/running/te_model_live_backlog.md` ;
7. update `Training Results Master Summary.md` when the accepted current best status or campaign interpretation changes.

Closeout Decision

`TE Curve Verification Pipeline` remains closed as the current official offline verification report. The accepted baseline for future work is unchanged:

- forward paper-derived comparison: `paper_retuned_best_Fw` ;
- backward paper-derived comparison: `paper_retuned_best_Bw` ;
- strongest individual paper-reference family evidence:
`rcim_retuned_GBM19_Fw` and `rcim_retuned_GBM19_Bw` ;
- strongest repository-owned static baseline: `tree` / `hist_gradient_boosting` . The `wave 2.1` temporal branch is accepted as verified exploratory evidence, not as the promoted offline winner. The practical next branch should either tune a compact temporal/hybrid model against the `tree` baseline or move the strongest verified candidates into deployment-readiness checks before any TwinCAT-facing promotion decision.